MECH 467 Project III: Simulation of Contouring Performance in Coordinated Two Axis Motion

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1 ABSTRACT

The objective of this lab is to become familiar with trajectory generation using trapezoidal feedrate profiling, to become familiar with designing lead-lag integrator controllers with various phase margin requirements, and to become familiar with modeling and analyzing the tracking and contouring errors resulting from these different trajectories and controllers. Finally, the students were to see this theoretical work occur in practice on a linear motor drive XY table.

2 INTRODUCTION

Multi-axis trajectories are heavily relied upon in industry, as machines are capable of doing many things to a degree of precision that humans are physically incapable of reaching unassisted. Specific tasks include machining parts and tools with reliable and repeatable geometries, or aligning pieces relative to each other during automated assembly processes.

This lab report consists of three parts. In part A the trajectory generation of the reference toolpath is described and discussed. Part B describes the design of three different lead-lag integrator controller: low bandwidth, high bandwidth, and mixed dynamics. The frequency data and transient response characteristics of these three controllers are discussed. Finally, part C investigates the contouring performance simulation work done to investigate the performance of the three controllers from part B with the trajectory generated in part A. Performance is assessed for different bandwidths, different feedrate speeds, and mismatched X and Y controller characteristics. This section ends with a comparison of the trajectory obtained during lab testing and the simulations from earlier, as well as a brief discussion on the self-designed trajectory that was also physically tested on the linear motor drive XY table in the lab.

3 Part A – Trajectory Generation

In part A a series of x and y commands were generated to produce the reference toolpath to be used in the lab. These were generated using trapezoidal feedrate profiling. Figure 1 shows the x axis position commands plotted against the y axis position commands.

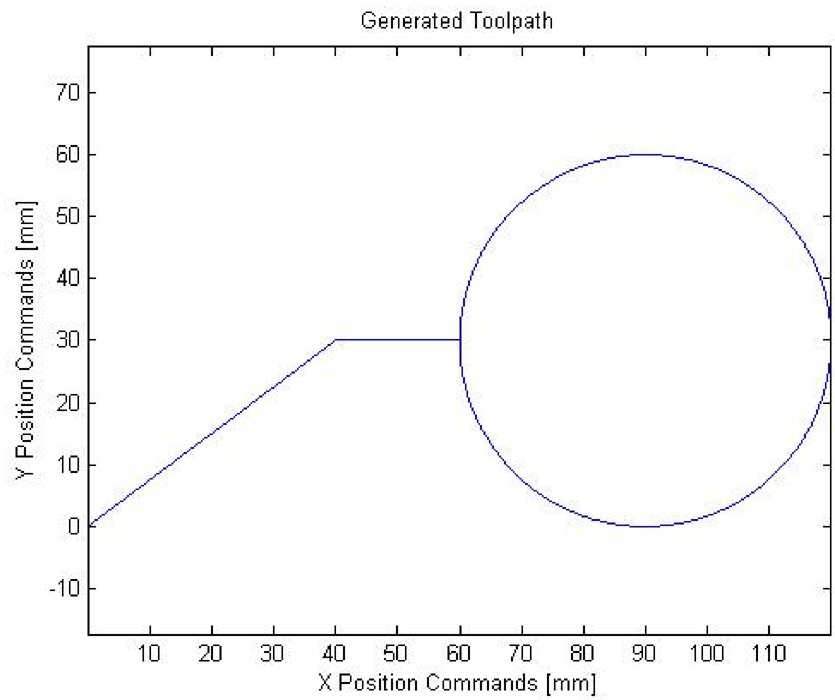

Figure 1: Generated Toolpath

The displacement (s), feedrate ($\dot{s}$), and tangential acceleration ($\ddot{s}$) profiles are shown in Figure 2 as a function of time.

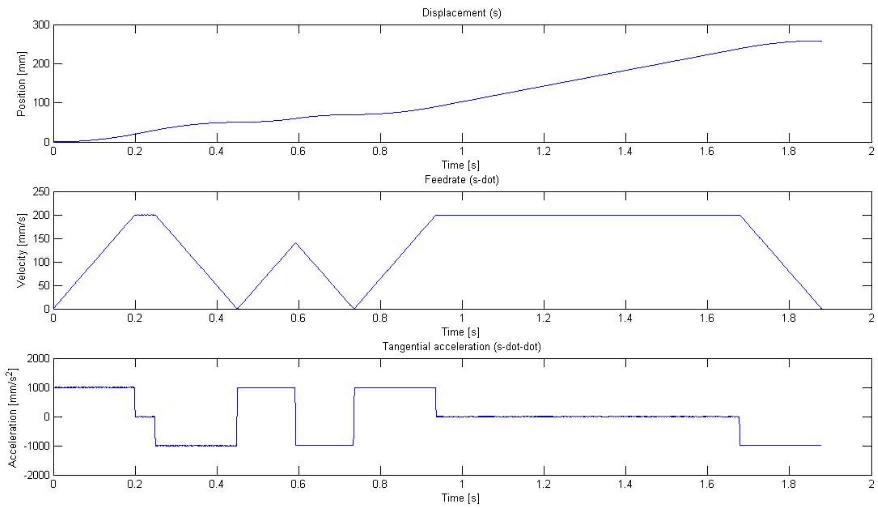

Figure 2: Displacement, Feedrate, and Tangential Acceleration Profiles

Note that the velocity plateaus near the maximum desired feedrate of 200 mm/sec. Because this trajectory is generated in discrete time it is not guaranteed to occur at exactly 200 mm/sec, but may occur at a slightly lower value. The feedrate decreases to zero between each segment, one of the lab requirements.

The resolved components of the displacement, feedrate, and tangential acceleration profiles shown in figure2 are the axis position, velocity and acceleration in each of the x and the y directions. These are shown in figure 3.

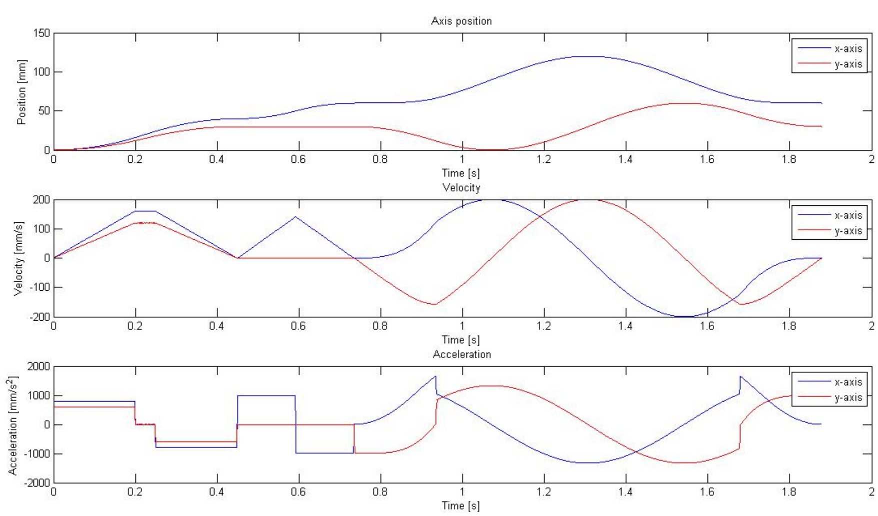

Figure 3: Axis Position, Velocity, and Acceleration Profiles

Unlike the displacement plot, the axis position is conservative – it shows actual position independent of path taken.

The tangential acceleration and deceleration limits only apply to tangential acceleration (figure 2), not axial acceleration (figure 3). Excessive axial acceleration could become a problem if it reaches the linear motor limits, and should be considered when limiting allowable user input to the machine, however it is not a consideration in this project. Axial accelerations higher than 1000 mm/sec^2 are allowed.

4 PART B – TWO AXIS CONTROLLER DESIGN

In part B three axial controllers were designed: a low bandwidth (LB) controller, a high bandwidth (HB) controller, and a mismatched dynamics (MD) controller. Since the MD controller is composed of a low bandwidth controller on the Y axis and a high bandwidth controller on the X axis, it is often discussed at the same time as the LB or HB controller in the following discussion. In this section the controller transfer functions are presented, along with some bode plots for the controllers. Finally, poles, zeros, system bandwidth, and the transient responses of the different controllers are discussed. A hand derivation of the transformation of the low bandwidth controller from continuous to discrete time is shown in Appendix A.

4.1 TRANSFER FUNCTIONS

The X and Y axis transfer functions of the lead-lag integrator controllers are shown in Table 1, in the continuous time domain.

Table 1: X and Y Continuous Time Domain Transfer Functions

	X	Y
LB	0.02408 s^2 + 1.156 s + 10.73 ---------------------------- 0.002245 s^2 + s	0.1045 s^2 + 5.057 s + 47.04 --------------------------- 0.002267 s^2 + s
HB	0.04814 s^2 + 4.538 s + 83.64 ---------------------------- 0.001094 s^2 + s	0.209 s^2 + 19.77 s + 365 ------------------------ 0.0011 s^2 + s
MD	0.04814 s^2 + 4.538 s + 83.64 ---------------------------- 0.001094 s^2 + s	0.1045 s^2 + 5.057 s + 47.04 --------------------------- 0.002267 s^2 + s

The MD X axis transfer function is the same as the HB X axis transfer function, and the MD Y axis transfer function is the same as the LB Y axis transfer function. This was expected, as mentioned in this introductory blurb to this section. Because of this, the MD results will often be combined with LB or HB results in the following sections.

The continuous transfer functions were converted to the discrete time domain using the c2d matlab command, with Tustin's method of time discretization. The resulting discrete transfer functions for the X and Y axis are shown in Table 2.

Table 2: X and Y Discrete Time Domain Transfer Functions

	X	Y
LB	8.982 z^2 - 17.54 z + 8.561 --------------------------- z^2 - 1.636 z + 0.6358	38.69 z^2 - 75.54 z + 36.87 --------------------------- z^2 - 1.639 z + 0.6387
HB	31.63 z^2 - 60.37 z + 28.79 --------------------------- z^2 - 1.373 z + 0.3728	136.9 z^2 - 261.2 z + 124.5 --------------------------- z^2 - 1.375 z + 0.375
MD	31.63 z^2 - 60.37 z + 28.79 --------------------------- z^2 - 1.373 z + 0.3728	38.69 z^2 - 75.54 z + 36.87 --------------------------- z^2 - 1.639 z + 0.6387

Once again the MD transfer functions are the same as the LB and HB transfer functions for Y and X respectively.

The X and Y axis transfer functions for the LB controller were converted to the discrete time domain by hand using a backwards Euler approximation. The discrete transfer functions found using this method were slightly different from those found using the matlab c2d command. This is hypothesized to be because the Tustin method is implemented differently from the Euler method. They were similar enough that the Tustin results were deemed appropriate for use in the rest of this lab.

4.2 Bode Plots

Although not required as part of this lab, some bode plots for the controllers are presented here, as it is interesting to see the final controller bode plot evolve from the components that contribute to producing it.

The lead-lag integrator controller is produced by cascading an integrator controller with a lead-lag controller designed to meet the given phase margin requirements. This is shown for the X and the Y axis in Figure 4.

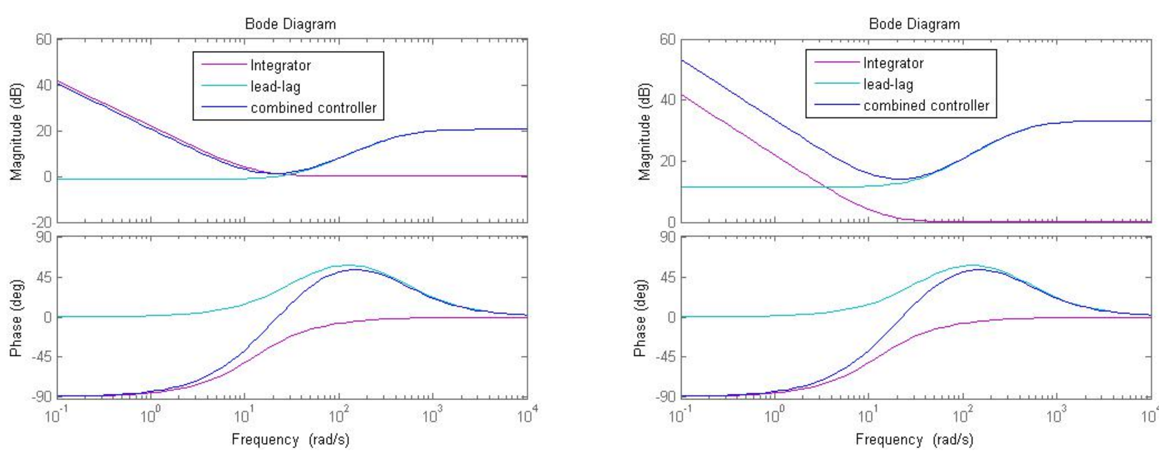

Figure 4: Low Bandwidth Lead-Lag Integrator Bode for X Axis (left) and Y Axis (right)

The combined controller is composed of a lead-lag controller and an integrator controller. The integrator controller is identical between axis, hence the pink trace in the two graphs in Figure 4 are identical. The phase contributed by the integrator controller is negligible. This is why the integrator controller has the formula (Ki + s)/ s – the plus s term on the top negates the 90 degrees phase contribution from the s on the bottom. Ki could be chosen to be even smaller if even this small amount of phase contribution is considered too high. The gain contributed by the integrator is high for low frequencies. This was discussed in lab 2, as it is what causes the integrator to respond so quickly, resulting in a "ka-chunk" noise when the machine executes a step response.

The lead-lag controllers are different between axis, as phi-m value depends on the open loop plant system, and alpha and T depend on phi-m. The most noticeable difference between the X and Y controllers due to the different plants is a positive vertical offset of the Y axis lead-lag controller. Since the Y axis plant has higher mass and higher friction, it makes sense that the Y axis controller would need a larger magnitude gain to make the system move.

Figure 5 shows the LB and HB bode plots for just the X axis.

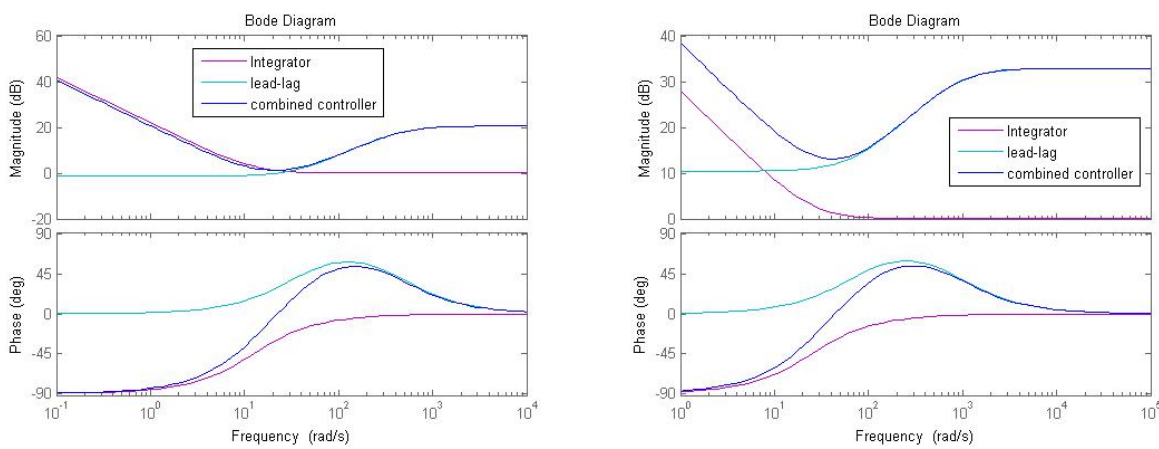

Figure 5: Low (left) and High (right) Bandwidth Lead-Lag Integrator Bode for X Axis

For the HB controller, the integrator constant Ki was different from that of the LB controller. Phi-m was also different between the two scenarios, hence T and alpha were also different. The HB controller bode plot is not shifted, since the DC gain is approximately 40 dB for both, but rather scaled, as the minimum gain for the HB controller is much higher than for the LB controller (approx. 13 dB instead of 0 dB). In addition, the frequency at which the maximum phase angle occurs for the HB controller is approximately double that of the LB controller (approximately 200 rad/s instead of 100 rad/s). This is expected, as this was part of the design – the high bandwidth controller phase margin occurs at a cross over frequency that is double that of the low bandwidth controller.

4.3 Frequency Data

The poles, zeros, and bandwidth of the closed loop system for each of the controllers are presented in Table 3.

Table 3: Frequency Data

	LB X Axis	LB and MD Y Axis	HB and MD X Axis	HB Y Axis
Poles	1.0e+02 * -2.7308 + 0.0000i -0.8173 + 0.2910i -0.8173 - 0.2910i -0.1214 + 0.0000i	1.0e+02 * -2.6716 + 0.0000i -0.8283 + 0.2853i -0.8283 - 0.2853i -0.1217 + 0.0000i	1.0e+02 * -5.7648 + 0.0000i -1.5826 + 0.6066i -1.5826 - 0.6066i -0.2410 + 0.0000i	1.0e+02 * -5.7057 + 0.0000i -1.5923 + 0.6023i -1.5923 - 0.6023i -0.2413 + 0.0000i
Zeros	-35.4575 -12.5664	-35.8067 -12.5664	-69.1261 -25.1327	-69.4741 -25.1327
Bandwidth [rad/s]	205.3534	205.3867	410.5319	410.5664

For a given controller the X and Y data is very similar. The slight differences can be attributed to the different X and Y plant parameters. The HB poles, zeros, and bandwidth are all approximately double those of the LB controller. This is expected, as the HB controller was designed to have a cross over frequency double that of the LB controller.

4.4 Transient Response Data

The transient closed loop responses for each axis are shown in Table 4.

Table 4: Transient Response Data

	LB X Axis	LB and MD Y Axis	HB and MD X Axis	HB Y Axis
Rise time [s]	0.0087	0.0087	0.0043	0.0043
Settling time [s]	0.0674	0.0662	0.0339	0.0339
Overshoot [%]	26.9820	26.5016	27.6677	27.5458
Undershoot [%]	0	0	0	0
Peak [mm]	1.2593	1.2576	1.2639	1.2630
Peak time [s]	0.0246	0.0245	0.0123	0.0123

Similar to the findings associated with Table 3 for the frequency data, the differences between X and Y data for a given controller are slight. For the speed of response attributes (rise time, settling time, and peak time) the HB controller responds about twice as fast as the LB controller. The HB controller does have more overshoot – this can be considered a trade-off for the fast response time. Once again, these differences can be attributed to the cross over frequency resulting in bandwidth for the HB controller that is twice that of the LB controller.

5 Part C – Contouring Performance Simulation

In part C the three controllers from part B were used in Simulink to model the system performance when faced with the toolpath generate din part A. Tracking and contour error were compared for the different controllers.

5.1 Simulated Tool Motion

For the three different controllers, the simulation predicted motion paths were laid over top of the reference toolpaths (Figure 6, 7, and 8).

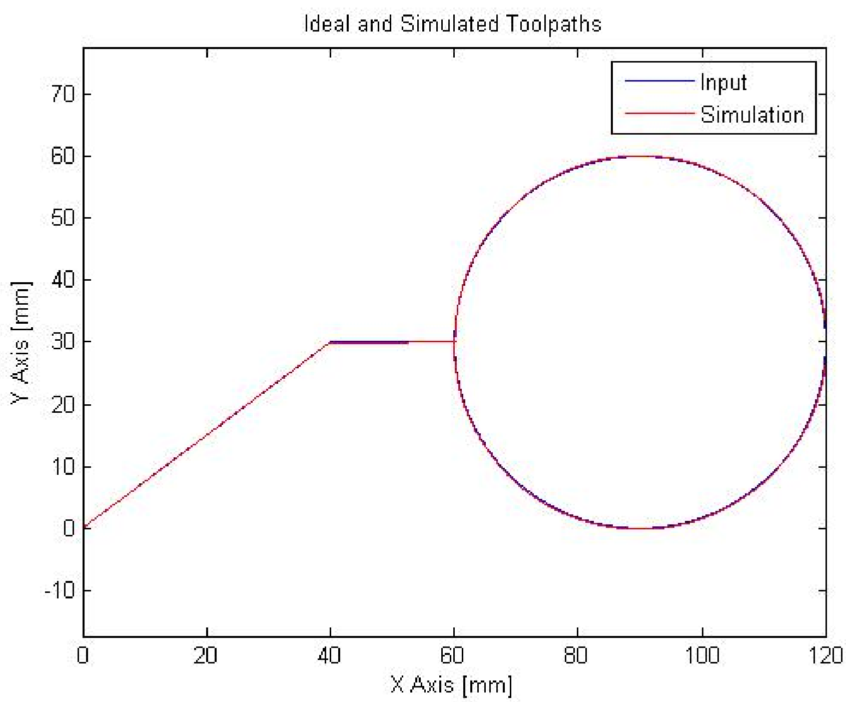

Figure 6: Simulated and Reference Toolpaths with Low Bandwidth Controller

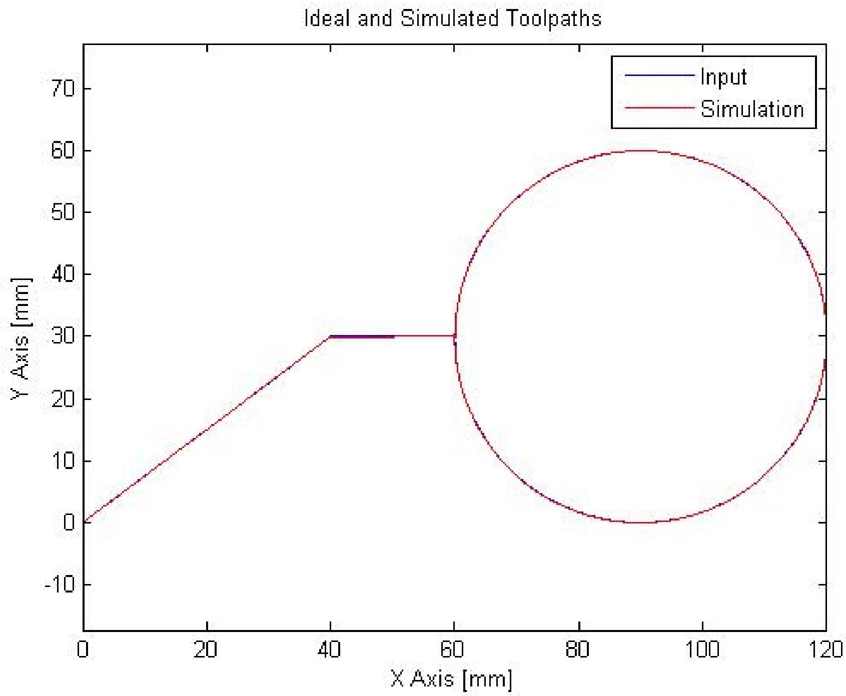

Figure 7: Simulated and Reference Toolpaths with High Bandwidth Controller

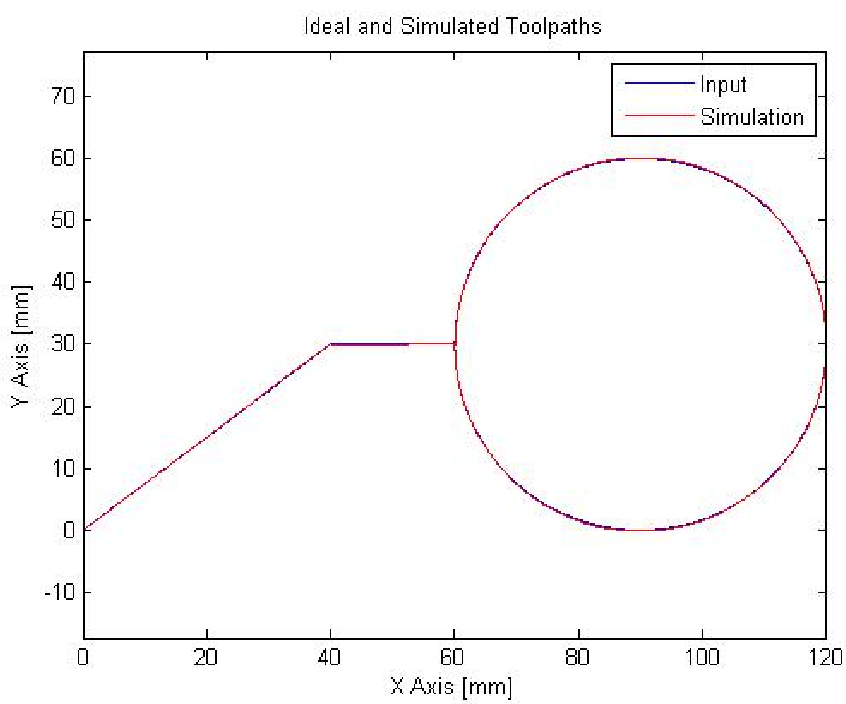

Figure 8: Simulated and Reference Toolpaths with Mixed Dynamics Controller

The simulated paths follow the macro reference profile. By visual inspection, deviations may occur

- Along the diagonal line for the mixed dynamics controller,
- Along the first half of the horizontal line for all three controllers,
- Around the majority of the circle for the low bandwidth controller, and
- At the lower and upper extremes of the mixed dynamics controller.

It is reasonable that the high bandwidth controller follows the reference profile most faithfully, as the presence of higher frequencies in the response region means the system can respond better to small changes in x and y, unlike the low bandwidth system where the system “freezes” as it can’t respond as quickly as is desired.

For the mixed dynamics system, the x axis has the high bandwidth controller and the y axis has the low bandwidth controller. This explains why the circle trace only deviates from the desired path at the top and the bottom – the x axis is capable of meeting the desired position whereas the y axis response is not as faithful to the desired trace. Additionally, this is the only controller that had visible errors in the diagonal line. This was not an issue for the low bandwidth controller, as both the x and the y controller were equally sluggish in their responds and therefore produced a straight line that followed the desired trajectory. However since the controllers for the MD controller are not matched this is no longer the case and errors become visible.

5.2 Tracking errors

Tracking error is the difference between the points in the reference toolpath vector and the points in the simulated response vector. It is time dependent, unlike contouring error, which is only the geometric difference between the machined and reference paths, regardless of where specific points occur. The tracking error can be visualized by observing the reference toolpath and the simulated toolpath as a function of time. It can be more accurately represented as the difference between these two traces.

Figure 9 shows the x axis trajectory profile for the LB and the HB and MD controllers, and figure 10 shows the tracking error for the LB and the HB and MD controllers for the x axis.

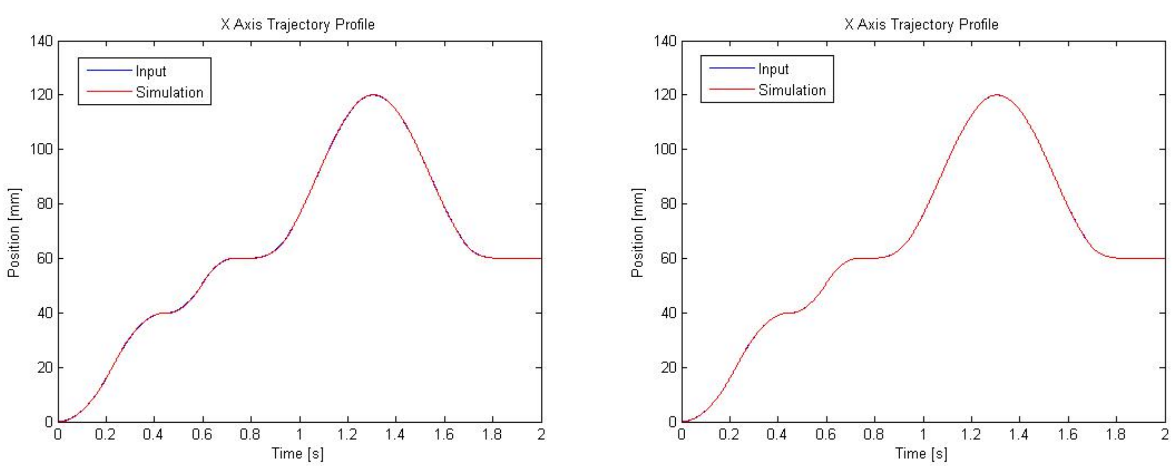

Figure 9: X Axis Trajectory Profile for LB controller (left) and HB and MD controllers (right)

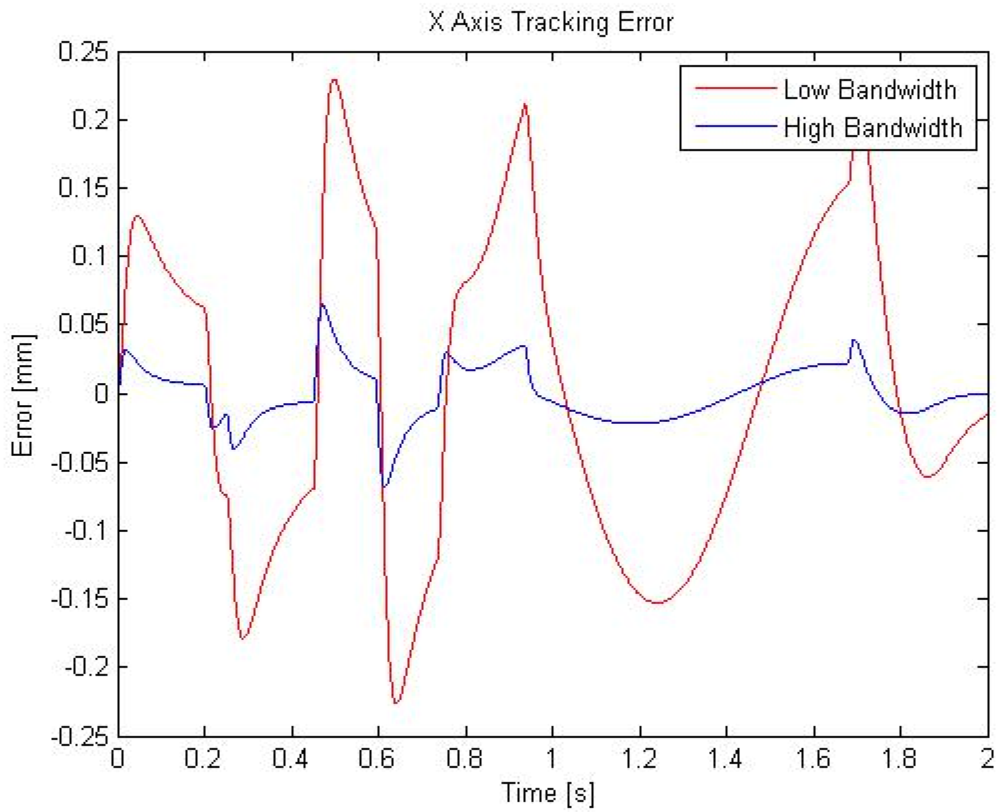

Figure 10: Tracking error for LB (red) and HB and MD controllers (blue)

Once again the high bandwidth response agrees with the reference trace much better than the low bandwidth controller does. Again this is due the high bandwidth resulting in quicker system response to match the reference trace. This is particularly evident in the tracking error graph – the errors from the high bandwidth system are at least three times smaller than those from the low bandwidth system.

The tracking errors are all positive (reference toolpath at that time is larger than simulated response) for sections of the trajectory with a positive derivative and negative for sections with a negative derivative. In a physical sense this means that the response takes a bit more time to get started (it is trying to keep up with the reference profile through the acceleration region) and eventually matches the reference curve during the constant acceleration region. When the reference curve starts decelerating it takes the simulated response some time to respond similarly.

Figure 11 shows the y axis trajectory profile for the LB and MD and for the HB controllers, and figure 12 shows the tracking error for the LB and MD, and HB controllers for the y axis.

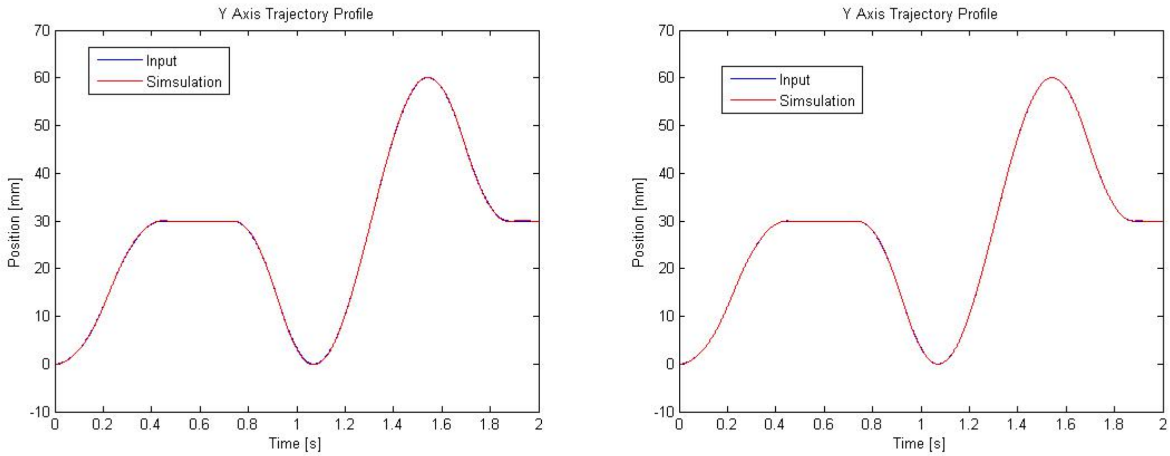

Figure 11: Y Axis Trajectory Profile for LB and MD controllers (left) and HB controller (right)

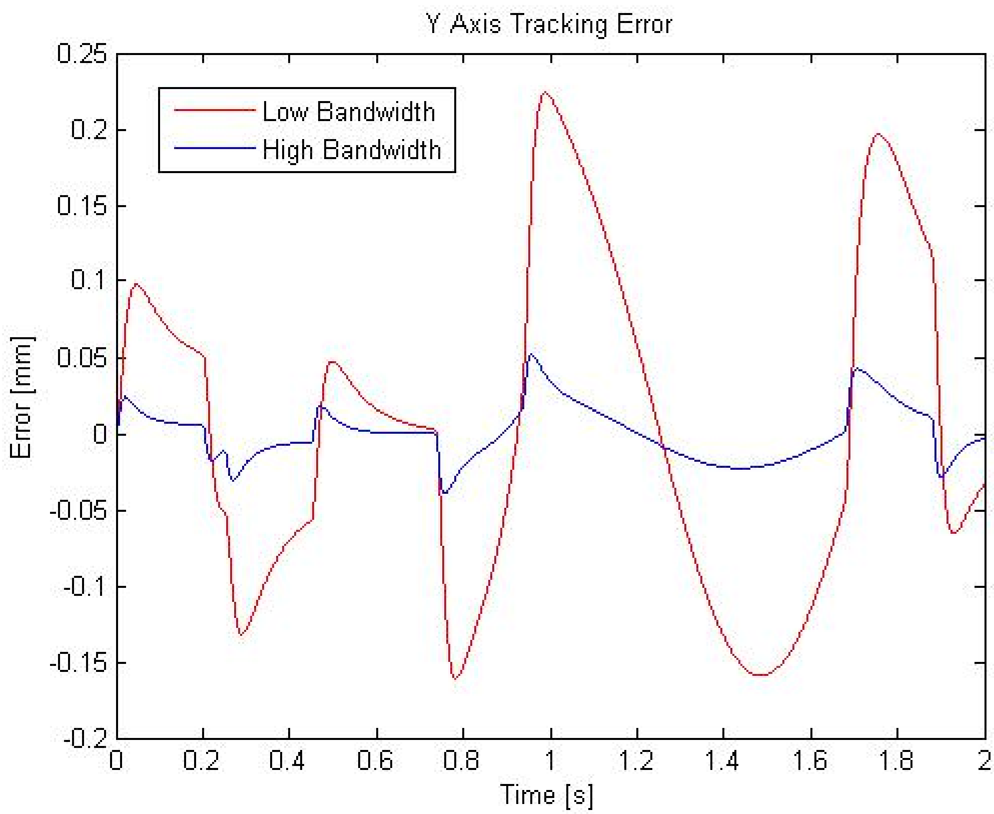

Figure 12: Tracking error for LB and MD (red) and HB controllers (blue)

Once again the simulated response has better agreement with the reference trajectory for the HB case. The tracking error is as much as 4 times as large for the LB as for the HB controller. Again the sign of the tracking error generally corresponds to the sign of the slope at a given point.

5.3 Contouring Error at Critical Regions

By looking at the agreement between simulated and reference toolpath traces values for the contouring error were estimated at the four critical regions identified in the lab manual. The results are presented in Table 5. The maximum, minimum, and half-way tracking errors in X and Y are also presented for critical region 1. The half-way tracking error was taken at 0.225 seconds – the time at which the reference path was half way from P1 to P2.

Table 5: Contour and [Tracking X, Tracking Y] Errors

	LB	HB	MD
R1	0.006 mm; [0.1294, 0.09801] mm max [-0.179, -0.1318] mm min [-0.04307, -0.02675] mm half	0.0006 mm; [0.03162, 0.02383] mm max [-0.04084, -0.03047] mm min [-0.02483, -0.01792] mm half	0.006 mm [0.03161, 0.09799] mm max [-0.04084, -0.1317] mm min [-0.02483 , -0.02675] mm
R2	0.06 mm	0.005 mm	0.05 mm
R3	0.12 mm	0.013 mm	0.012 mm
R4	0.15 mm	0.016 mm	0.1 mm

As expected the HB contouring errors were much smaller than the LB contouring errors – they were an order of magnitude smaller everywhere. The MD controller contouring error agreed with that of the LB controller at R1, R2, and R4. R3 was at the end of a long segment with only X axis movement, therefore the R3 contouring error agreed with that of the HB controller.

The agreement between the LB and the MD contouring error at R1 is surprising, as the hypothesis had been that despite the worse tracking error of the LB controller, the contouring error should still be quite small as the X and Y controllers responded with the same dynamics and therefore should have produced a faithful representation of the straight line, albeit slightly sluggishly. This appears not to be the case as the MD controller contouring error is just as bad as the LB contouring error.

5.4 Increased Feedrate: F = 250 mm/sec for LB Controller

The reference toolpath was regenerated with an increased feedrate of 25 mm/sec. The updated dynamic profiles are shown in Figures 13 and 14.

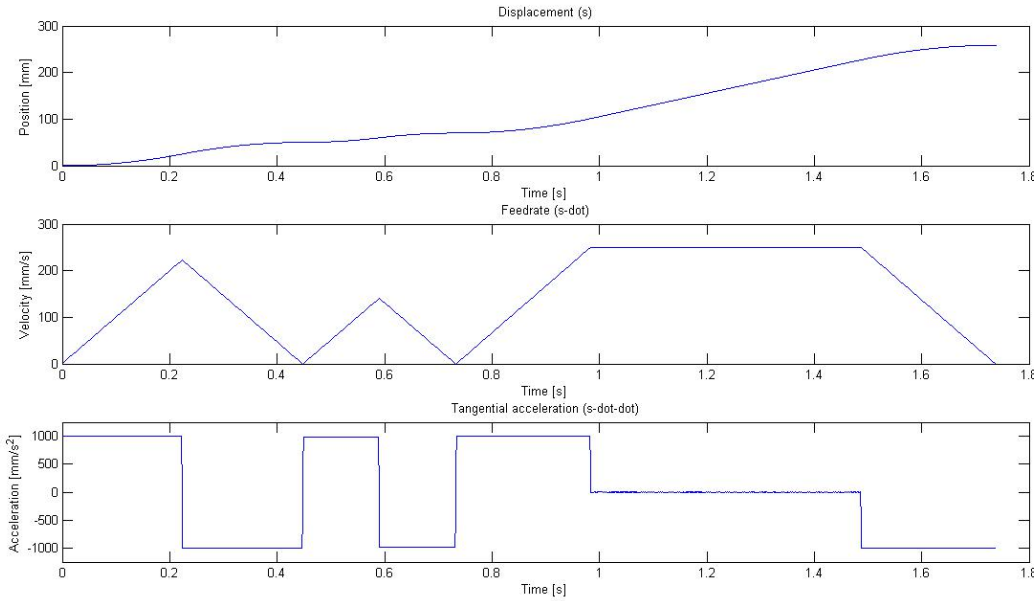

Figure 13: Displacement, Feedrate, and Tangential Acceleration Profiles with F = 250 mm/s

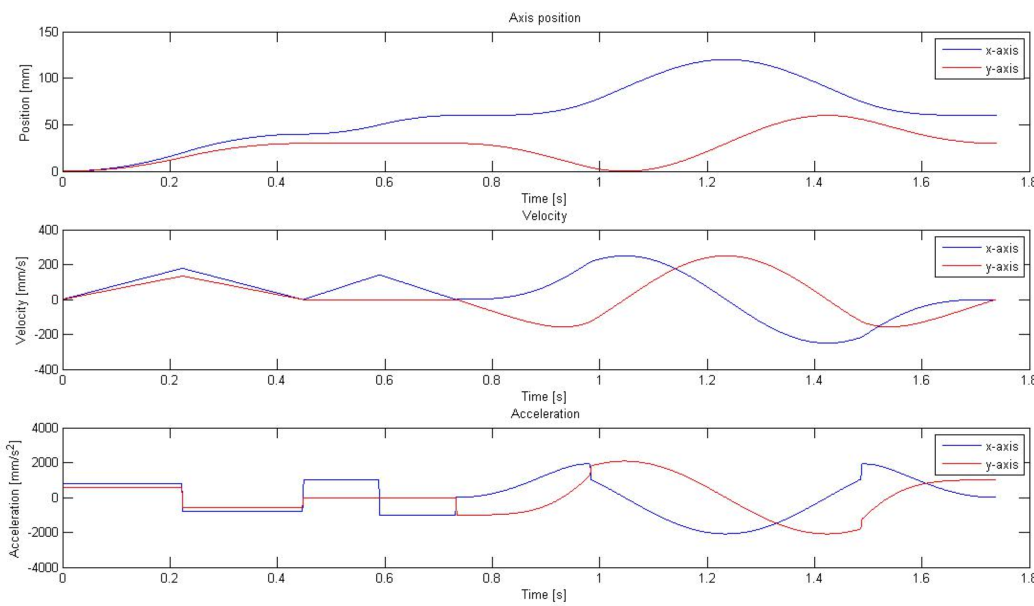

Figure 14: Displacement, Feedrate, and Tangential Acceleration Profiles with F = 250 mm/s

The resulting toolpath generated with the LB controller is shown next to the original low velocity toolpath in figure 15.

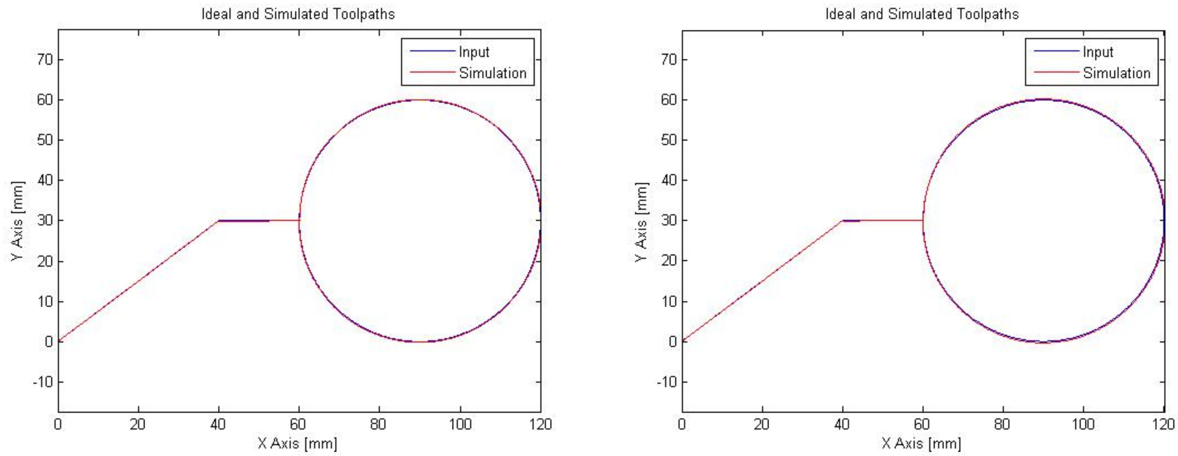

Figure 15: Simulated and Reference Toolpaths with Low Bandwidth Controller for Low (left) and High (right) Feedrates

The discrepancy between the reference and the simulated paths is much more evident in the high speed scenario. At a higher speed the demands on the controller to respond quickly are even more stringent and therefore emphasize the inability of the controller to faithfully reproduce the reference trace.

The X axis trajectory profiles of the low and the high feedrate cases and their tracking errors are shown for the LB controller in figures 16 and 17.

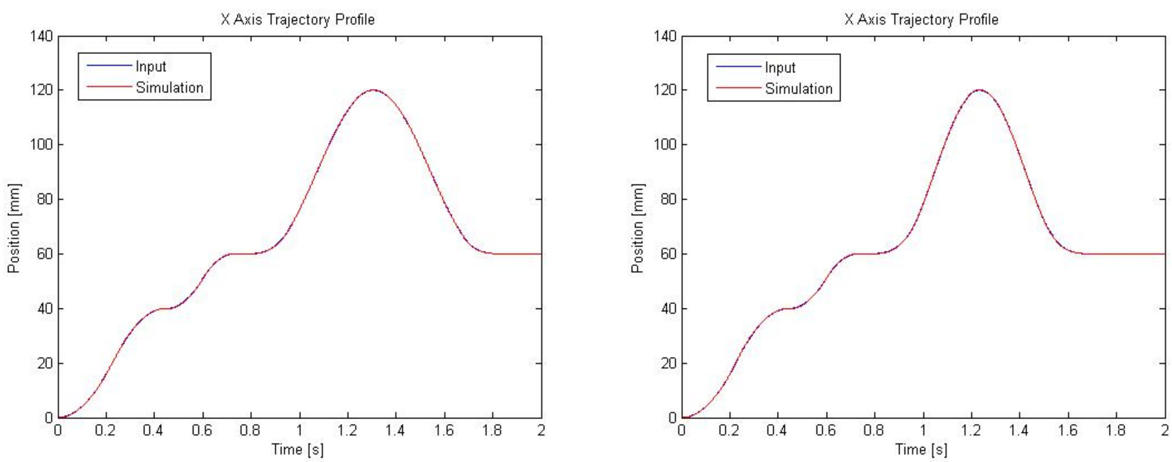

Figure 16: X Axis LB Controller Trajectory Profile for low (left) and high (right) Feedrates

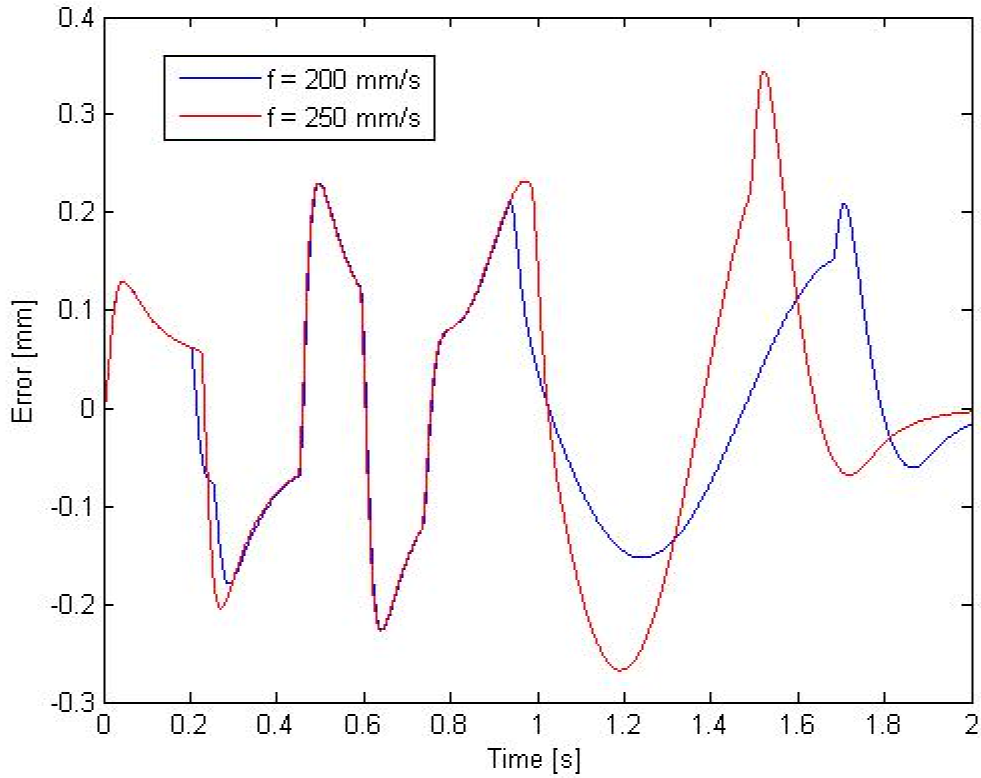

Figure 17: Tracking Error for LB X Axis for Low (blue) and High (red) Feedrates

From the trajectory profiles the tracking errors between the low and high feedrates appear tiny. Looking at the tracking error plot in Figure 17 the segments where maximum feedrate was not reached are identical whereas sections where maximum feedrate was reached have more extreme tracking errors – the maximums are higher and the minimums are lower. This is especially evident in the circular trajectory portion around t = 1.2s where the tracking error is more extreme. The travel is more condensed too, as expected, as the peaks from 1.2s onwards occur earlier in time for the high velocity profile than for the low velocity profile.

Figures 18 and 19 also show the trajectory profile and tracking error for the LB controller low vs high feedrates for the Y axis.

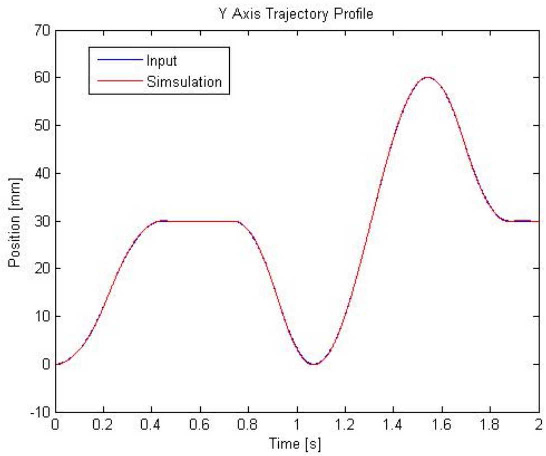

Figure 18: Y Axis LB Controller Trajectory Profile for low (left) and high (right) Feedrates

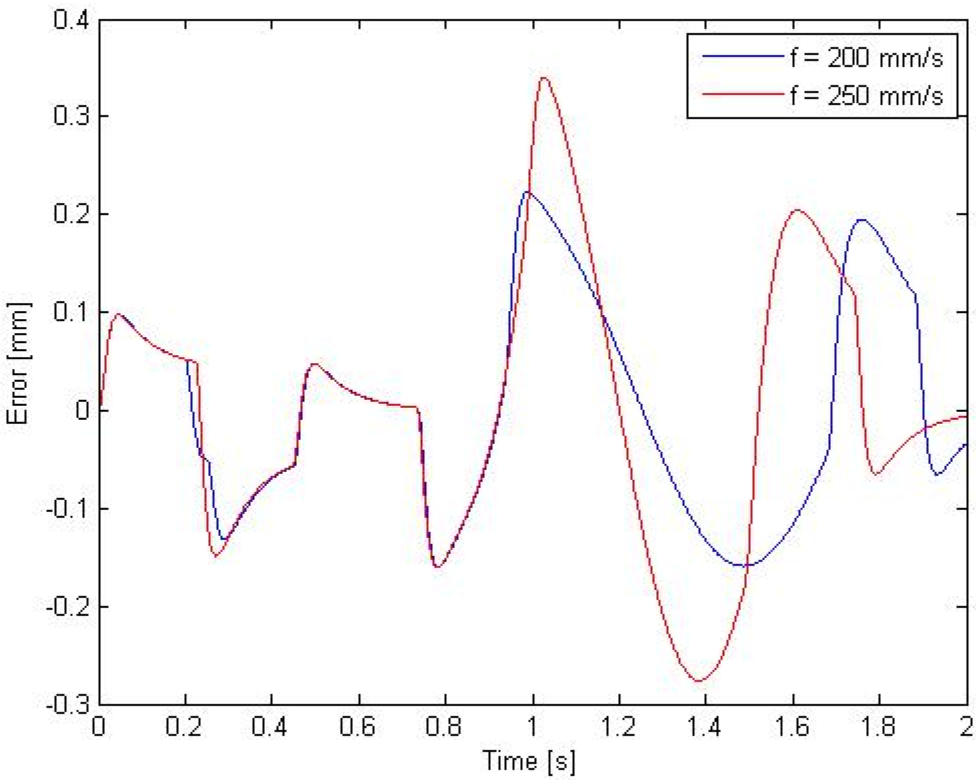

Figure 19: Tracking Error for LB Y Axis for Low (blue) and High (red) Feedrates

The observations for the X axis again apply for the Y axis. The major difference between tracking errors of low and high feedrates is evident in the circle in the diagram. The errors are emphasized in the higher feedrate scenario, and the extremes are more extreme.

The contouring error difference between the low and high feedrate scenarios for the LB controller are shown in Table 6.

Table 6: Contour and [Tracking X, Tracking Y] Errors for Low and High Feedrate LB Controller

	F = 200 mm/s	F = 250 mm/s
R1	0.006 mm; [0.1294, 0.09801] mm max [-0.179, -0.1318] mm min [-0.04307, -0.02675] mm half	0.004 mm; [0.1289, 0.09767] mm max [-0.2042, -0.149] mm min [-0.05743, -0.04885] mm half
R2	0.06 mm	0.06 mm
R3	0.12 mm	0.12 mm
R4	0.15 mm	0.25 mm

The Contouring error in the R1, R2, and R3 regions was approximately the same between different feedrate scenarios. It was almost double in the R4 region however, as was expected from observations that the error was more pronounced around the circular region.

Throughout this lab report it has been repeated that controllers are only capable of responding so fast. Earlier this came through as larger bandwidth being required to faithfully trace circles of a given radius; in this section it comes through as controllers only being capable of faithfully tracing circles at certain speeds. In the end it all comes down to the bandwidth dictating the angular velocity that the controller is capable of reproducing. Angular velocity is equal to feedrate divided by radius, so an increase in feedrate or decrease in radius will make it more difficult for the controller to faithfully reproduce the circle, while a decrease in feedrate or increase in radius relax the bandwidth requirements of the controller.

5.5 Increased Feedrate: F = 250 mm/sec for MD Controller

The high feedrate was also used in comparing the MD controller trajectory. Figure 20 shows a comparison of the MD controller toolpaths for low and high feedrates.

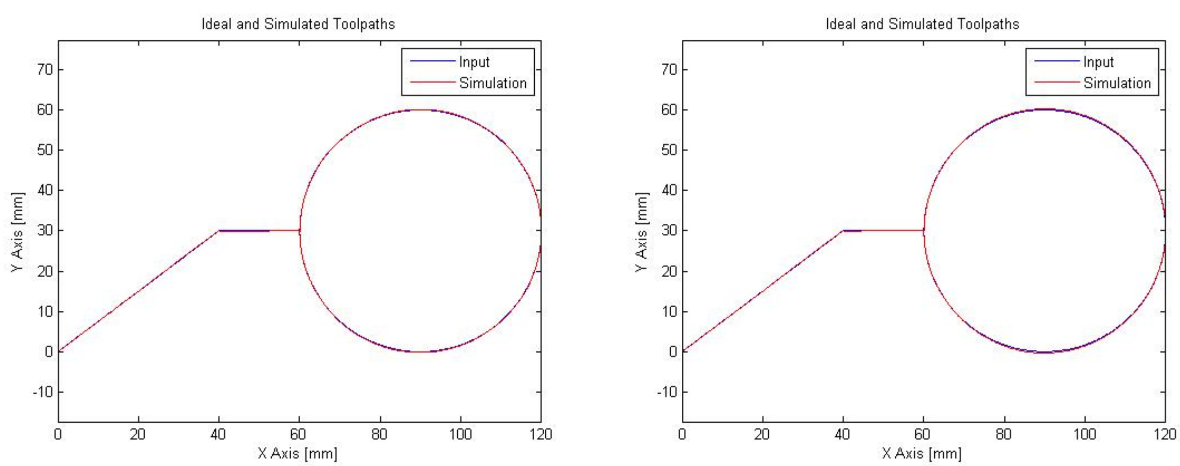

Figure 20: Simulated and Reference Toolpaths with Mixed Dynamics Controller for Low (left) and High (right) Feedrates

Keeping in line with the MD controller having lower agreement with the reference trajectory at the top and bottom of the circle, and the increased feedrate emphasizing discrepancies between reference and simulated trajectories, the errors visible at the top and bottom of the circle in the higher feedrate case are more pronounced than in the lower feedrate case.

The contouring errors between the low and high feedrate cases for the MD controller are shown in Table 7.

Table 7: Contour and [Tracking X, Tracking Y] Errors for Low and High Feedrate MD Controller

	F = 200 mm/s	F = 250 mm/s
R1	0.006 mm [0.03161, 0.09799] mm max [-0.04084, -0.1317] mm min [-0.02483 , -0.02675] mm	0.04 mm [0.0315, 0.047] mm max [-0.05682, -0.1491] mm min [0.00549 , 0.04885] mm
R2	0.05 mm	0.05 mm
R3	0.012 mm	0.013 mm
R4	0.1 mm	0.2 mm

The contour errors for R2, R3, and R4 are approximately the same between the low and the high feedrate case. The error at R1 however, is much higher for the high feedrate case.

Recall that following the comparison of contouring errors for all three controllers at the lower feedrate, a note was made indicating the agreement between the LB and the MD contouring error for region R1 was surprising. Here the error has increase by a magnitude with increased speed, while in Table 2 we did not see this error increase for the LB controller. This new information reaffirms that as long as both axis have a similar bandwidth and therefore respond similarly to each other, even if that trajectory profile does not perfectly agree with the reference profile it will yield a product with low contouring error, as the axis are equally sluggish and end up following the desired trajectory. When the axis have different characteristics they do not respond in agreement with each other and the errors of the slower axis become more pronounced and have a more detrimental effect on the final result.

5.6 Lab Results – LB Controller with Line-Circle Trajectory

In the lab the line-circle trajectory was loaded onto the XY table software, and the machine traced out the image. The actual table position information was recorded. Figure 21 shows a comparison between the simulated and the actual toolpaths.

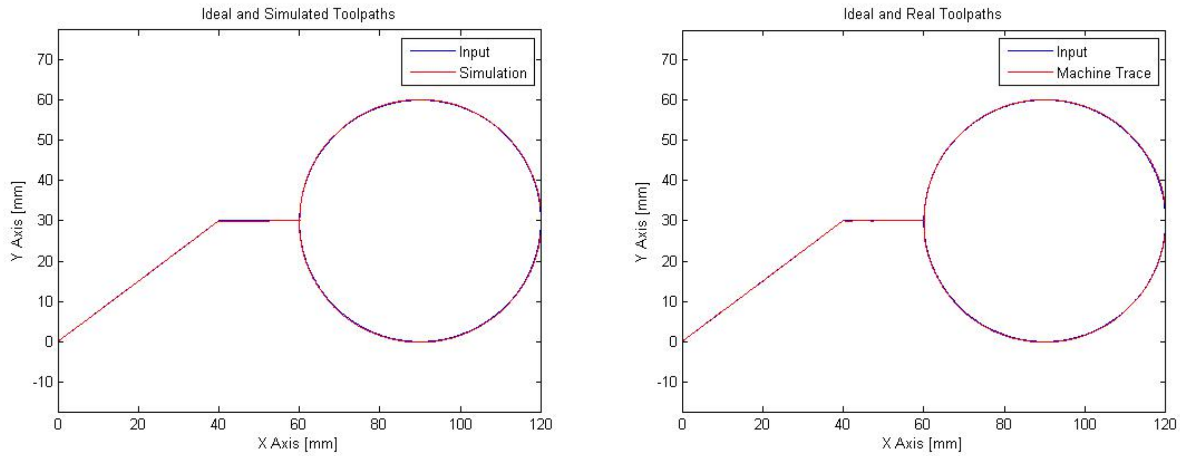

Figure 21: Ideal and Simulated (left) and Actual (right) Toolpaths for Low Bandwidth Controller with Low Feedrate

It appears as though there is less contouring error at the start of the horizontal line, and around the circle than was expected from the simulation.

The X and Y axis trajectory profiles are shown in Figure 22.

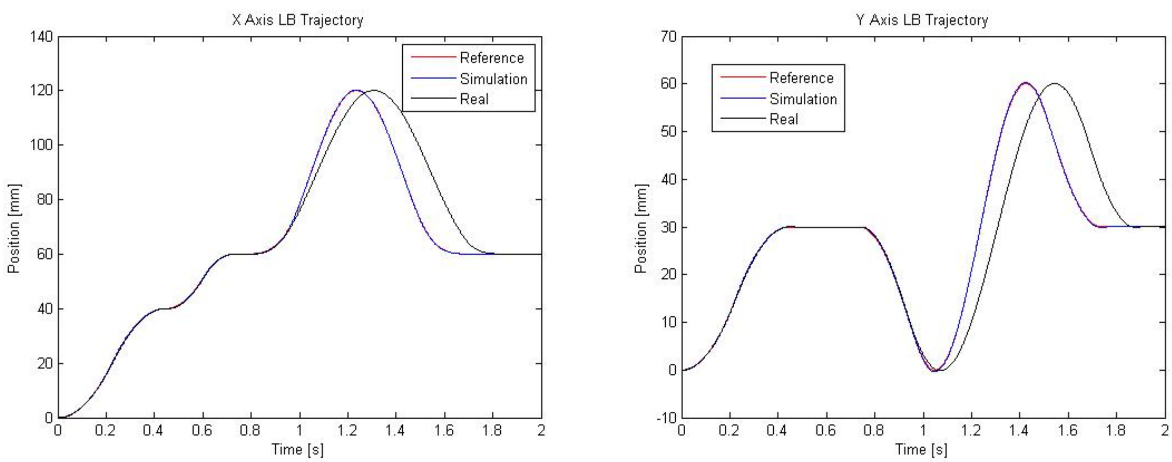

Figure 22: X Axis (left) and Y Axis (right) Trajectory Profiles for LB controller with Simulated and Real Trajectory Data

There appears to be good agreement between the simulation and the real trajectory until the circle is traced out, at which point the real trajectory lags behind the simulation. This may be due to the presence of friction, which was not included as an input in the simulation used to model the system behaviour.

By inspecting the plot of the Real and Reference toolpath shown in figure 21, contouring data was determined for the real case. This information is compared with the simulated data for the LB case, shown in Table 8.

Table 8: Contour Errors for Low Feedrate LB Controller Simulated and Real Data

	Simulated Response	Real Response
R1	0.006 mm;	0.006 mm;
R2	0.06 mm	0.05 mm
R3	0.12 mm	0.08 mm
R4	0.15 mm	0.15 mm

The contouring error measured from the real response curve is almost identical to that of the simulation. It only deviated at R3, where the real response was much closer to the reference trajectory than the simulated response was. This is the point where the real trajectories plotted in figure 22 began to lag severely behind the simulated ones; hence the presence of friction may have slowed the machine trajectory enough that it was about to come to a halt quicker at the end of the horizontal line in the real situation than it had in the simulation.

5.7 Self-Designed Trajectory

In the final section of the lab, each student designed their own trajectory to implement on the XY table. My design consisted of several long and short lines, and several large and small circles (Figure 23).

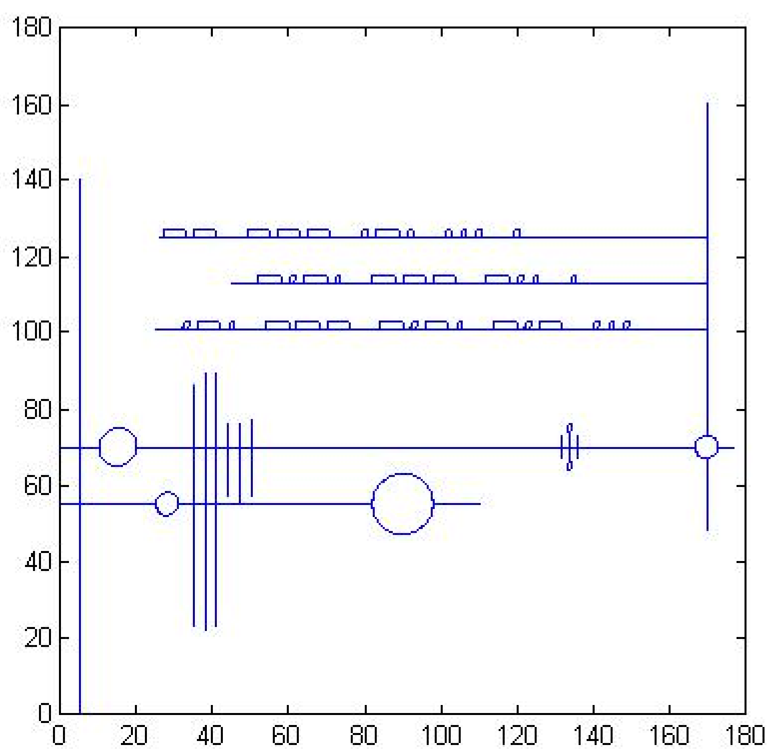

Figure 23: Self-Designed Trajectory

The circles were drawn with the acceleration and deceleration values set to 30 and -30 mm/s^2, as the smaller radius circles caused very high X and Y accelerations with the lab-specified acceleration limits of ±250 mm/s^2.

The first two times it was attempted on the table, it failed at the beginning of the Morse Code section. Upon investigation this appeared to be due to a jump caused by restating the same x1 to x2 commands twice. It is interesting to note that the machine's behaviour upon finding this impossible command initially appeared random and chaotic, but turned out to be very repeatable as the second time it drew the same trajectory, complete with the same vertical oscillations causing light and dark pencil marks along the chaotic traces.

Once the jump was removed the drawing executed with no problems. The machine's ability to draw circles of 1mm radius with high fidelity was very impressive.

6 CONCLUSION

In conclusion, the trapezoidal profiling method provides a reasonable, albeit simplistic trajectory for a linear actuator to follow. The sharp acceleration changes result in infinite jerk which is non-ideal for the system, but for applications that don't require extreme precision it is adequate.

The high bandwidth controller has the best response characteristics. Applications with high feedrates and/ or small radii require a high bandwidth controller, whereas low bandwidth controllers are "good enough" for applications with both low feedrates and high circle radii.

Matched dynamics are always desirable, as they can still produce reasonable representations of the reference trace, whereas when the dynamics are mismatched the X and Y axis are no longer working together and therefore produce larger contouring errors. It is better overall to decrease the performance of the "better" controller to match the more limited one.

Finally, it was concluded that simulations can provide a very accurate representation of how the actual controller will perform with a given trajectory, however the more complete a model is the better the simulation will be. Friction plays a large role in affecting the physical behaviour of a plant.